Cluster values at every interior point of the complex ℓ^p ball

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Abstract

Every value of a character over an interior point of the complex ℓ^p ball is a weak cluster value in the same ball, for $1 < p < \infty$. The exact tail-radius estimate and origin approximation are proved directly. The conclusion concerns exactly the catalogue target; no historical priority claim is made.

Catalogue scope. AIM 673; complete resolution. Canonical target at revision `1e56463`. The interior-point question is recorded in [1, Question 8]; the corrigendum is explicitly accounted for below.

1 Statement and argument

Let $1 < p < \infty$, let $X = \ell^p(\mathbb{N}; \mathbb{C})$, and let B be its open unit ball. We prove the cluster-value equality

$$\mathrm{Cl}_B(f, z) = \{\varphi(f) : \varphi \in M_z\} \qquad (f \in A_u(B),\ z \in B). \tag{1}$$

Here M_z consists of the characters of $A_u(B)$ whose restriction to X^* is evaluation at z, and the cluster set uses weakly convergent nets in the same open unit ball. This includes the nonzero interior points and non-Hilbert exponents in the catalogue question. The mechanism is an exact estimate for a character on polynomials depending only on the tail coordinates.

We use continuous polynomials on a Banach space in their usual sense: finite sums of continuous homogeneous polynomials. They need not depend on finitely many coordinates. Their restrictions are dense in $A_u(B)$; indeed, radial dilations of a uniformly continuous holomorphic function converge uniformly to it on B, and the Taylor series of each strict radial dilation converges uniformly there. Every character of a unital uniform algebra has norm one.

The cluster-value theorem at the origin was proved in [2, Theorem 3.1]. We include the polynomial form needed below, so that the proof does not depend on any unproved general cluster-value theorem.

Lemma 1.1 (Origin approximation)**.** *Let $R > 0$ and let ψ be a character of $A_u(RB)$ such that $\psi(x^*) = 0$ for every $x^* \in X^*$. For every continuous polynomial Q on X, every weak neighborhood V of 0, and every $\varepsilon > 0$, there exists $v \in RB \cap V$ with*

$$|Q(v) - \psi(Q)| < \varepsilon. \tag{2}$$

The same statement holds for any coordinate-tail copy of ℓ^p.

*The solutions and partial results in this document were found by AI. The supplied research and the separate repository review are AI work; human peer review and formal verification are not claimed.

Proof. Write $T_K x = (0, \dots, 0, x_{K+1}, x_{K+2}, \dots)$. Because $X^* = \ell^{p/(p-1)}$, a sufficiently large K satisfies $RB_{T_K X} \subset V$. The polynomial identity

$$Q(x) - Q(T_K x) = \sum_{j=1}^{K} x_j Q_j(x)$$

holds for suitable continuous polynomials Q_j. It follows either by multilinear expansion or by freezing the coordinates one at a time. Consequently $\psi(Q \circ T_K) = \psi(Q)$. If (2) failed throughout $RB_{T_K X}$, then $|Q(T_K x) - \psi(Q)| \geq \varepsilon$ for all $x \in RB$. The function

$$x \longmapsto \frac{1}{Q(T_K x) - \psi(Q)}$$

would then be holomorphic and uniformly continuous on RB, and hence belong to $A_u(RB)$. It would invert an element whose value under ψ is zero, contradicting multiplicativity. □

Lemma 1.2 (Exact tail radius)**.** *Fix $\varphi \in M_z$ and a positive integer N. Split $X = H_N \oplus_p Y_N$ into the first N coordinates and the remaining coordinates, and put*

$$z = z_H + z_T, \qquad u = \|z_H\|_p, \qquad r = (1 - u^p)^{1/p}. \tag{3}$$

For every continuous polynomial Q on Y_N,

$$|\varphi(Q \circ T_N)| \leq \sup_{\|y\|_p < r} |Q(y)|. \tag{4}$$

Thus $Q \mapsto \varphi(Q \circ T_N)$ extends to a character η_N of $A_u(rB_{Y_N})$, lying in the fiber over z_T.

Proof. First let Q be k-homogeneous with $k \geq 1$, and set $\|Q\|_1 = \sup_{\|y\|_p < 1} |Q(y)|$. If $u > 0$, choose a complex-linear functional ℓ on H_N of norm one with $\ell(z_H) = u$. Put

$$c = \frac{k u^{p-1}}{1 - u^p}.$$

The function $x \mapsto e^{c\ell(x_H)} Q(x_T)$ belongs to $A_u(B)$. For $t = \|x_H\|_p$ and $x \in B$, its modulus is at most

$$\|Q\|_1 e^{ct} (1 - t^p)^{k/p}.$$

The logarithm of the last factor has derivative

$$c - \frac{k t^{p-1}}{1 - t^p}.$$

Since $t \mapsto t^{p-1}/(1 - t^p)$ is strictly increasing on $(0, 1)$, the maximum occurs at $t = u$. Therefore

$$\|e^{c\ell(x_H)} Q(x_T)\|_B \leq e^{cu} r^k \|Q\|_1.$$

On the other hand, multiplicativity, the uniformly convergent exponential series, and the fiber condition give

$$\varphi(e^{c\ell(x_H)} Q(x_T)) = e^{c\ell(z_H)} \varphi(Q(x_T)) = e^{cu} \varphi(Q(x_T)).$$

The norm-one bound for φ now yields

$$|\varphi(Q(x_T))| \leq r^k \|Q\|_1. \tag{5}$$

When $u = 0$, take $c = 0$ and $r = 1$ to obtain the same estimate. It also holds for constant polynomials, with $k = 0$.

Now let Q have degree d, and for an integer $m \geq 1$ decompose $Q^m = \sum_{k=0}^{md} Q_{m,k}$ into homogeneous components. Cauchy's coefficient formula for the one-variable polynomial $\lambda \mapsto Q(\lambda y)^m$ gives

$$r^k \|Q_{m,k}\|_1 = \|Q_{m,k}\|_{rB_{Y_N}} \leq \|Q^m\|_{rB_{Y_N}} = \|Q\|^m_{rB_{Y_N}}.$$

Using (5), linearity and multiplicativity, we obtain

$$|\varphi(Q(x_T))|^m \leq \sum_{k=0}^{md} |\varphi(Q_{m,k}(x_T))| \leq (md+1)\|Q\|^m_{rB_{Y_N}}.$$

Taking mth roots and letting $m \to \infty$ proves (4). Polynomial density extends this unital multiplicative functional continuously to $A_u(rB_{Y_N})$. Its value on a linear functional on Y_N is that functional's value at z_T, as claimed. □

Theorem 1.3. *The equality* (1) *holds for every* $1 < p < \infty$.

Proof. Fix $f \in A_u(B)$, $z \in B$, and $\varphi \in M_z$. It suffices to show that every weak neighborhood of z contains a point $w \in B$ with $f(w)$ arbitrarily close to $\varphi(f)$. We first give the construction for one fixed continuous polynomial P on X.

For each N, use (3) and write $t_N = \|z_T\|_p$ and $R_N = r + t_N$. On polynomials Q on Y_N, define

$$\psi_N(Q) = \eta_N(Q(\cdot - z_T)).$$

The exact tail-radius bound and the triangle inequality imply

$$|\psi_N(Q)| \leq \sup_{\|y\|_p < r} |Q(y - z_T)| \leq \sup_{\|v\|_p < R_N} |Q(v)|.$$

Hence ψ_N extends to a character of $A_u(R_N B_{Y_N})$. It is in the zero fiber because $\eta_N(L) = L(z_T)$ for all linear functionals L on Y_N.

Freezing the finite head coordinates of a polynomial gives an exact algebraic identity

$$P(x_H + x_T) - P(z_H + x_T) = \sum_{j=1}^{N} (x_j - z_j) P_j(x)$$

for continuous polynomials P_j. Applying φ yields

$$\varphi(P) = \eta_N(P(z_H + \cdot)).$$

Consequently the polynomial $Q_N(v) = P(z + v)$ on Y_N satisfies

$$\psi_N(Q_N) = \varphi(P). \tag{6}$$

The origin-approximation lemma supplies points $v \in R_N B_{Y_N}$ arbitrarily close weakly to 0 for which $P(z + v)$ is arbitrarily close to $\varphi(P)$.

These preliminary points have a precisely controlled norm excess:

$$\|z + v\|_p \leq L_N := (u^p + (r + 2t_N)^p)^{1/p}. \tag{7}$$

As $N \to \infty$, $t_N \to 0$, $u \to \|z\|_p$, and $r \to (1 - \|z\|_p^p)^{1/p}$, so $L_N \to 1$. Set $a_N = (L_N + 1/N)^{-1}$ and $w = a_N(z + v)$. Then $w \in B$ by (7), and $a_N \to 1$. Since a continuous polynomial is uniformly continuous on bounded sets, and the L_N are bounded,

$$\sup_{\|x\|_p \leq L_N} |P(a_N x) - P(x)| \longrightarrow 0. \tag{8}$$

This also follows directly by writing P as a finite sum of homogeneous polynomials and using $a_N^k \to 1$ for each degree k.

To check the weak topology explicitly, a prescribed neighborhood of z contains a set given by finitely many inequalities $|x_j^*(w-z)| < \delta$, $1 \le j \le m$. Choose N large enough that $|(a_N - 1)x_j^*(z)| < \delta/2$ for all j and that (8) is as small as desired. Then use the origin-approximation lemma to choose $v \in R_N B_{Y_N}$ with $|a_N x_j^*(v)| < \delta/2$ for all j, and with $|P(z+v) - \varphi(P)|$ as small as desired. The resulting $w = a_N(z+v)$ is in the prescribed weak neighborhood and in B, and $P(w)$ is arbitrarily close to $\varphi(P)$.

For the original function f, first choose one polynomial P with $\|f - P\|_B < \varepsilon$. Choose N and v as just described for this fixed polynomial, making $|P(w) - \varphi(P)| < \varepsilon$. Because $\|\varphi\| = 1$,

$$|f(w) - \varphi(f)| \le |f(w) - P(w)| + |P(w) - \varphi(P)| + |\varphi(P - f)| < 3\varepsilon.$$

Directing the choices by weak neighborhoods and positive tolerances proves $\varphi(f) \in \mathrm{Cl}_B(f, z)$. The opposite inclusion follows by taking a convergent subnet of evaluation characters in the compact spectrum: its restriction to X^* is evaluation at the weak limit z. □

Scope and relation to the corrigendum. The final approximating points lie in the original open unit ball. The proof uses the exact identity $\|x_H + x_T\|_p^p = \|x_H\|_p^p + \|x_T\|_p^p$ and therefore asserts the result for ℓ^p, not for arbitrary Banach spaces with an approximation property. The corrigendum [3] withdraws the general ball-algebra conclusion obtained from an incorrect bounded-set argument. No result from that withdrawn argument is used here: the exact-radius estimate is proved above, and the origin approximation is included with its elementary proof.

References

[1] D. Carando, D. García, M. Maestre and P. Sevilla-Peris, *On the spectra of algebras of analytic functions*, Contemporary Mathematics **561** (2012), 165–198, Question 8. https://doi.org/10.1090/conm/561/11114.

[2] R. M. Aron, D. Carando, T. W. Gamelin, S. Lassalle and M. Maestre, *Cluster values of analytic functions on a Banach space*, Math. Ann. **353** (2012), 293–303. https://doi.org/10.1007/s00208-011-0681-0; https://mate.dm.uba.ar/~slassall/clustervalues.pdf.

[3] D. Carando, D. Galicer, S. Muro and P. Sevilla-Peris, *Corrigendum to "Cluster values for algebras of analytic functions"*, Adv. Math. **329** (2018), 1307–1309. https://doi.org/10.1016/j.aim.2018.03.019.